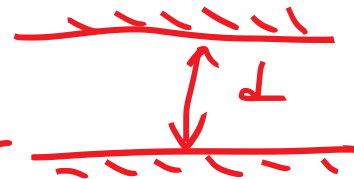


Parallel Plate Waveguide



$$\cos \theta_i = \frac{m\lambda}{2d}$$

$$\cos \theta_i = \frac{m\lambda}{2d}$$



$$\sin \theta_i = \sqrt{1 - \left(\frac{m\lambda}{2d}\right)^2}$$

$$\frac{m\lambda n}{d}$$

$$E_y = 2jE_{i0} e^{-j\beta \sqrt{1 - \left(\frac{m\lambda}{2d}\right)^2} z}$$

$$\sin(\beta n \cos \theta_i)$$

$$\vec{E}_y = 2jE_{i0} e^{-j\beta \sqrt{1 - \left(\frac{m\lambda}{2d}\right)^2} z} \sin\left(\frac{m\lambda n}{d}\right)$$

$$\vec{H}_z = -\frac{2E_{i0}}{\eta} e^{-j\beta \sqrt{1 - \left(\frac{m\lambda}{2d}\right)^2} z} \cos\left(\frac{m\lambda n}{d}\right) \left(\frac{m\lambda}{2d}\right)$$

$$\vec{H}_n = -\frac{2jE_{i0}}{\eta} e^{-j\beta \sqrt{1 - \left(\frac{m\lambda}{2d}\right)^2} z} \sin\left(\frac{m\lambda n}{d}\right) \sqrt{1 - \left(\frac{m\lambda}{2d}\right)^2}$$

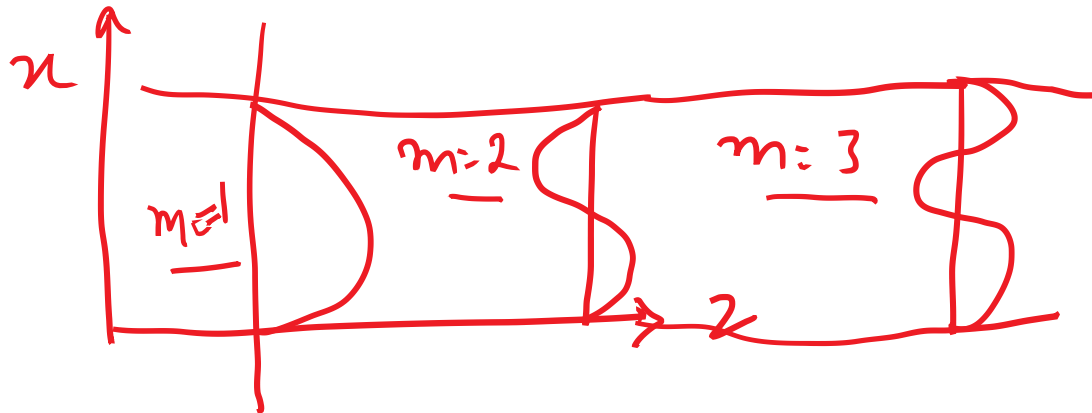
Parallel Plate Waveguide



$$\cos \theta_i = \frac{m\lambda}{2d}$$

$$\sin \theta_i = \sqrt{1 - \left(\frac{m\lambda}{2d}\right)^2}$$

What is the significance of m !
 $m=1, 2, 3, \dots$



Existence
of multi
modes

Parallel Plate Waveguide (Cut off wavelength/frequency)



$$\underline{e^{-j\beta \sin\theta_i z}}$$

$$\beta \sin\theta_i = \beta \sqrt{1 - \left(\frac{m\lambda}{2d}\right)^2}$$

Wave to propagate $\beta \sin\theta_i$ should be always
a real value

$$\left(\frac{m\lambda}{2d}\right) \leq 1$$

$$\lambda_c \leq \frac{2d}{m}$$

$$\text{So } \left[\begin{array}{l} \lambda < \lambda_c \\ f > f_c \end{array} \right]$$

For wave to
propagate

$$\text{So } \boxed{\lambda_c = \frac{2d}{m}}$$
$$\boxed{f_c = \frac{m v_p}{2d}}$$

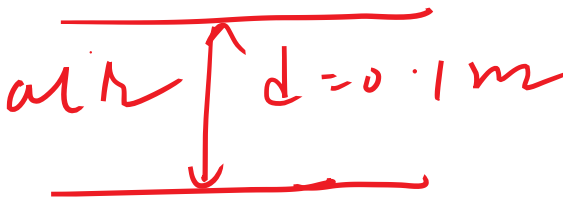
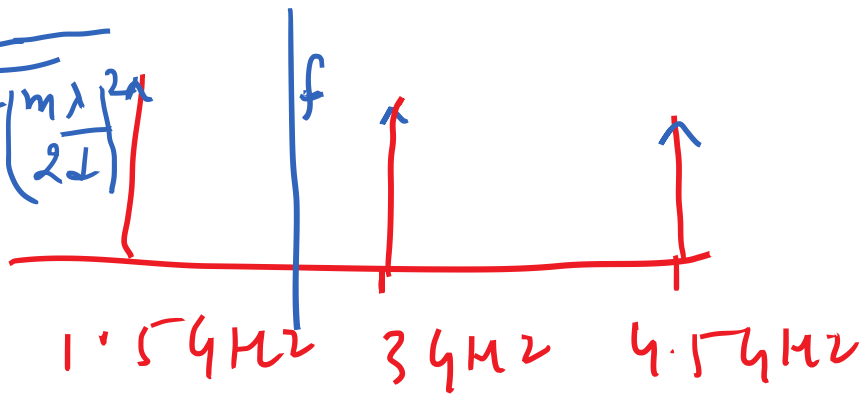
Parallel Plate Waveguide (Cut off wavelength/frequency)



Example

$$\beta \sin \theta_i = \frac{2\pi}{\lambda} \sqrt{\left(1 - \frac{m\lambda}{2d}\right)^2} = \beta$$

$$\lambda_g = \frac{\lambda}{\sqrt{1 - \left(\frac{m\lambda}{2d}\right)^2}}$$



$$f_c = \frac{m \times 3 \times 10^8}{2 \times 0.1}$$

$$f_c = m \cdot 1.5 \times 10^9 \text{ Hz}$$

$$f_{c1} = 1.5 \text{ GHz}$$

$$f_{c2} = 3 \text{ GHz}$$

$$f_{c3} = 4.5 \text{ GHz}$$

[fundamental mode is preferable.] → Dispersion

TE_m [since $E_z = 0$] It has H_n & H_z

Parallel Plate Waveguide (Cut off wavelength/frequency)



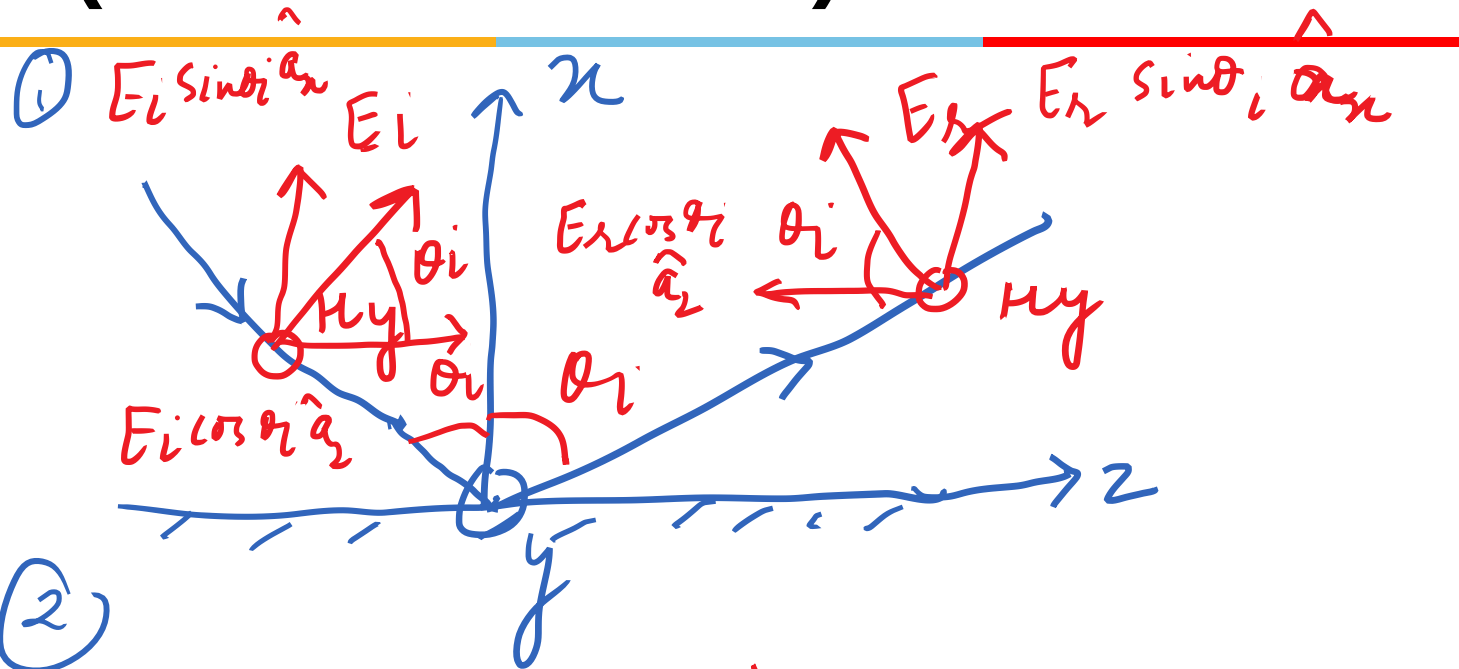
$$\left[\lambda_c = \frac{2d}{m} \right] \quad \left[f_c = \frac{m v_p}{2d} \right]$$

$$\left[\beta' = \beta \sqrt{1 - \left(\frac{m\lambda}{2d} \right)^2} = \beta \sqrt{1 - \left(\frac{m\lambda}{\lambda_c} \right)^2} \right]$$

$$\frac{2\pi}{\lambda_g} = \frac{2\pi}{\lambda} \sqrt{1 - \frac{\lambda^2}{\lambda_c^2}}$$

$$\left[\lambda_g = \frac{\lambda}{\sqrt{1 - \frac{\lambda^2}{\lambda_c^2}}} \right]$$

Parallel Plate Waveguide (TM Polarization)



②

$$\vec{H}_y = \frac{2E_{i0}}{\eta} \cos\left(\frac{m\pi x}{a}\right) e^{-j\beta z \sin\theta_i} \hat{a}_y$$

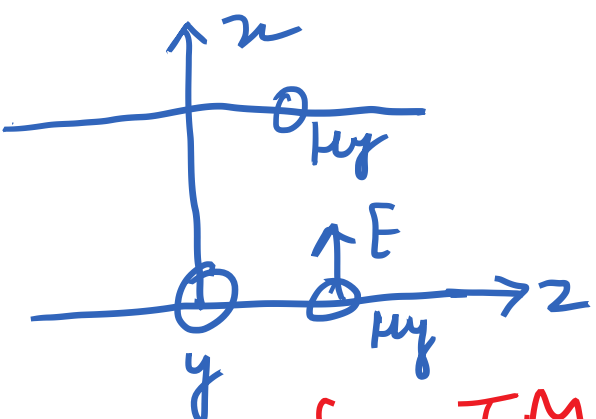
$$\vec{E}_x = 2E_{i0} e^{-j\beta z \sin\theta_i} \sin\theta_i \cos\left(\frac{m\pi x}{a}\right) \hat{a}_x$$

$$\vec{E}_z = +j2E_{i0} e^{-j\beta z \sin\theta_i} \cos\theta_i \sin\left(\frac{m\pi x}{a}\right) \hat{a}_z$$

TM₀

Parallel Plate Waveguide (TM Polarization)



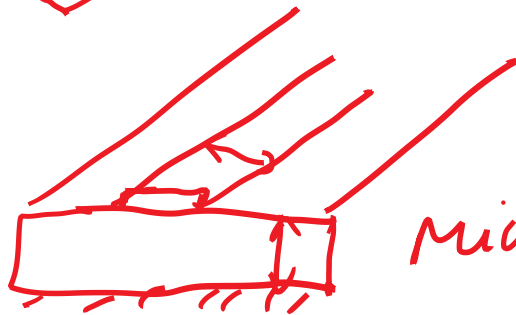
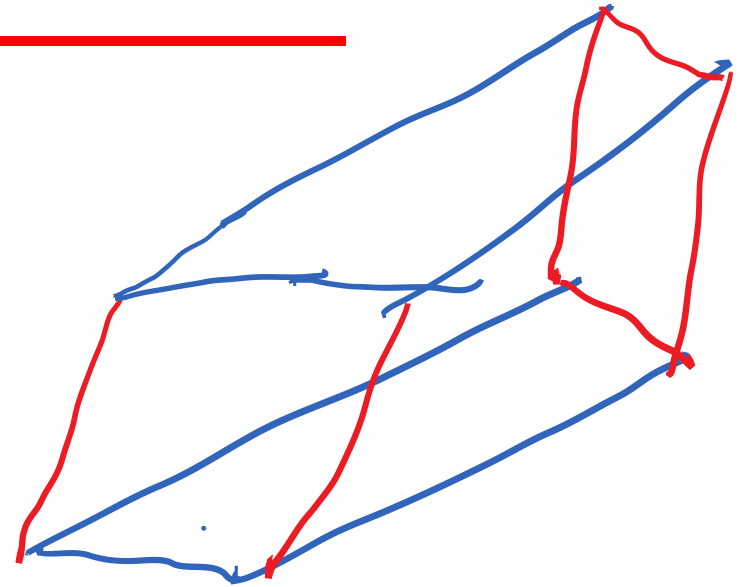
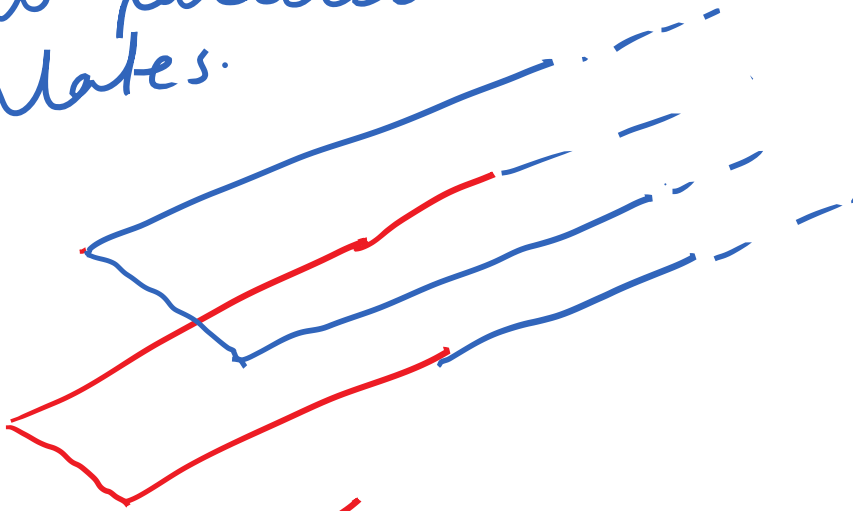
$TM_0 \rightarrow E_y \text{ exists}$ $H_z = 0$
 $H_n \text{ exists.}$

 $\hookrightarrow$ Direction of propagation
 So TM_0 exists $\approx$ TEM mode

whereas $TE_0 \rightarrow E_y = 0$ only H_z
 $H_n \neq 0$ exists
 So TE_0 mode
 doesn't exist.

Parallel Plate Waveguide to Rectangular Waveguide

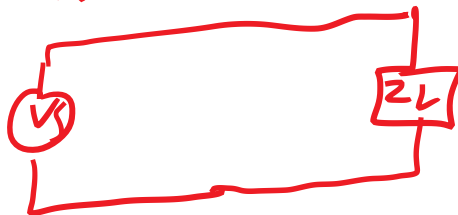


Two parallel plates.



microstrip line

Rectangular Waveguide



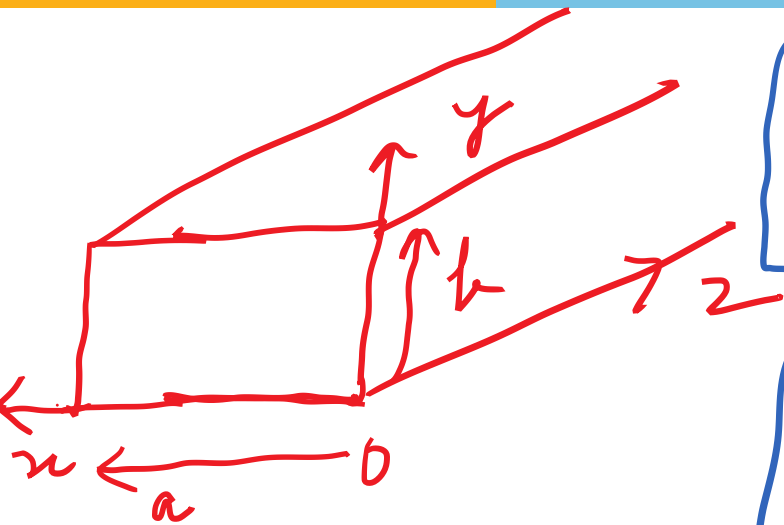
Transmission line

circular waveguide



Rectangular Waveguide

(TM)



$$a > b$$

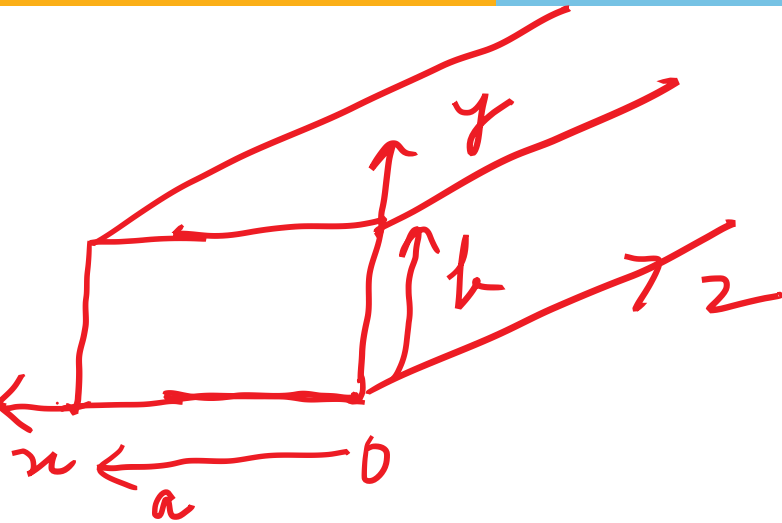
(E_z, H_z) are longitudinal components

(E_x, E_y, H_x, H_y) are transverse or normal components

TM $\rightarrow$ Transverse magnetic Mode

Rectangular Waveguide

(TM)



$$\gamma_{TM} = k_z = 0$$

$$\nabla^2 E_z = -\omega^2 \mu \epsilon E_z$$

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \frac{\partial^2 E_z}{\partial z^2} = -\omega^2 \mu \epsilon E_z$$

$$E = X Y Z$$

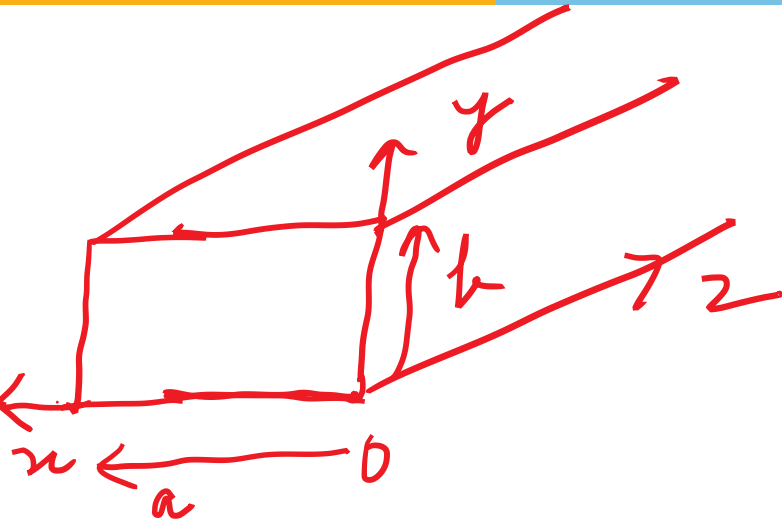
$$YZ \frac{\partial^2 X}{\partial x^2} + XZ \frac{\partial^2 Y}{\partial y^2} + XY \frac{\partial^2 Z}{\partial z^2} = -\omega^2 \mu \epsilon XYZ$$

$$\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} + \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} = -\omega^2 \mu \epsilon$$

Rectangular Waveguide

(TM)

$$\gamma_M = \mu_2 = 0$$



$$\frac{1}{x} \frac{\partial^2 x}{\partial x^2} = -A^2$$

$$\frac{1}{y} \frac{\partial^2 y}{\partial y^2} = -B^2$$

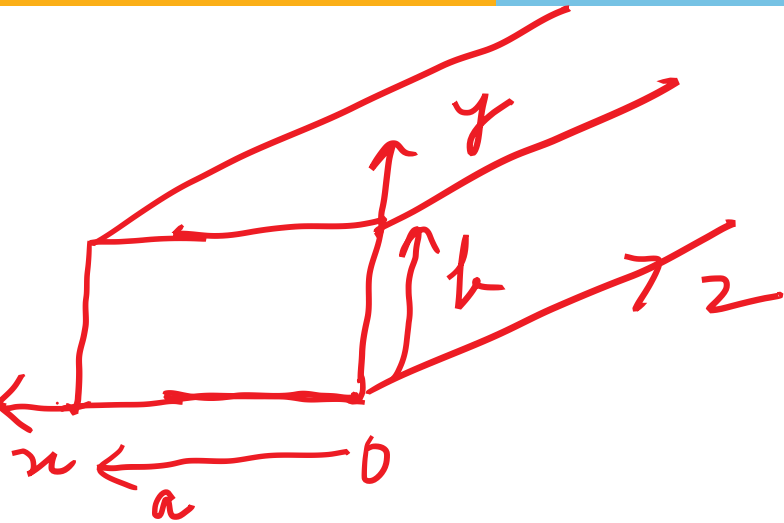
$$\frac{1}{z} \frac{\partial^2 z}{\partial z^2} = -\beta^2$$

$$\frac{1}{x} \frac{\partial^2 x}{\partial x^2} + \frac{1}{y} \frac{\partial^2 y}{\partial y^2} + \frac{1}{z} \frac{\partial^2 z}{\partial z^2} = \underline{-\omega^2 \mu \epsilon}$$

Rectangular Waveguide

(TM)

$$\gamma_{TM} = k_z = 0$$



$$\frac{1}{x} \frac{\partial^2 x}{\partial x^2} = -A^2$$

$$\frac{1}{y} \frac{\partial^2 y}{\partial y^2} = -B^2$$

$$\frac{1}{z} \frac{\partial^2 z}{\partial z^2} = -\beta^2$$

Along x & y dirⁿ, there is standing wave,
along z dirⁿ, there is travelling wave

Rectangular Waveguide

(TM)

$$\gamma_{TM} = k_z = 0$$

$$X = C_1 \cos Ax + C_2 \sin Ax \quad \leftarrow \quad \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = -A^2$$

$$Y = C_3 \cos By + C_4 \sin By \quad \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = -B^2$$

$$Z = C_5 e^{-j\beta' z} + C_6 e^{j\beta' z} \quad \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} = -\beta'^2$$

$$E_z = (C_1 \cos Ax + C_2 \sin Ax) (C_3 \cos By + C_4 \sin By) (C_5 e^{-j\beta' z} + C_6 e^{j\beta' z}) \cos \beta z$$

Rectangular Waveguide

(TM)

$$\gamma_{TM} = H_z = 0$$

$$E_z = (C_1 \cos Ax + C_2 \sin Ax) (C_3 \cos By + C_4 \sin By) (C_5 e^{-j\beta'z} + C_6 e^{j\beta'z}) \cos \beta z$$

Assume waveguide is infinite length

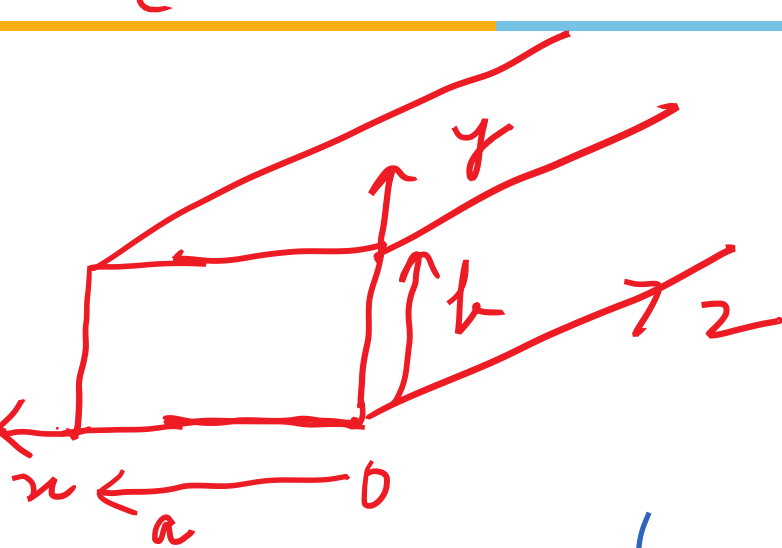
So No back wave

$$(C_6 = 0)$$

$$\text{So } E_z = e^{-j\beta'z} (C_1 \cos Ax + C_2 \sin Ax) (C_3 \cos By + C_4 \sin By) \cos \beta z$$

Rectangular Waveguide

(TM)



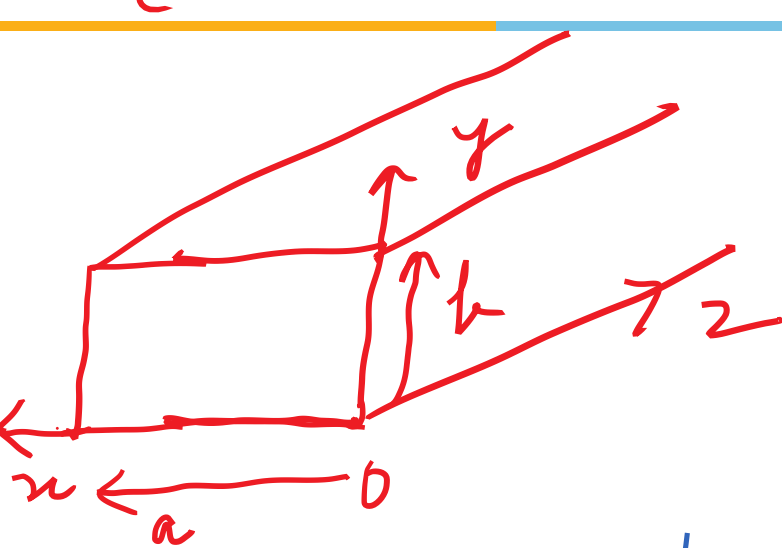
At $x=0$ $E_z = 0$
 means $C_1 = 0$
 Boundary condition

$$E_z = C_5 e^{-j\beta' z} (C_1 \cos Ax + C_2 \sin Ax) (C_3 \cos By + C_4 \sin By)$$

$$E_z = C_5 e^{-j\beta' z} C_2 \sin Ax (C_3 \cos By + C_4 \sin By)$$

Rectangular Waveguide

(TM)



$$\text{At } \underline{x=a} \quad E_z = 0$$

means

$$Aa = m\pi$$

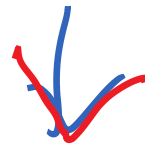
$$A = \frac{m\pi}{a}$$

$$E_z = C_5 e^{-j\beta'_z z}$$

$$C_2 \sin An (C_3 \cos By + C_4 \sin By)$$

So

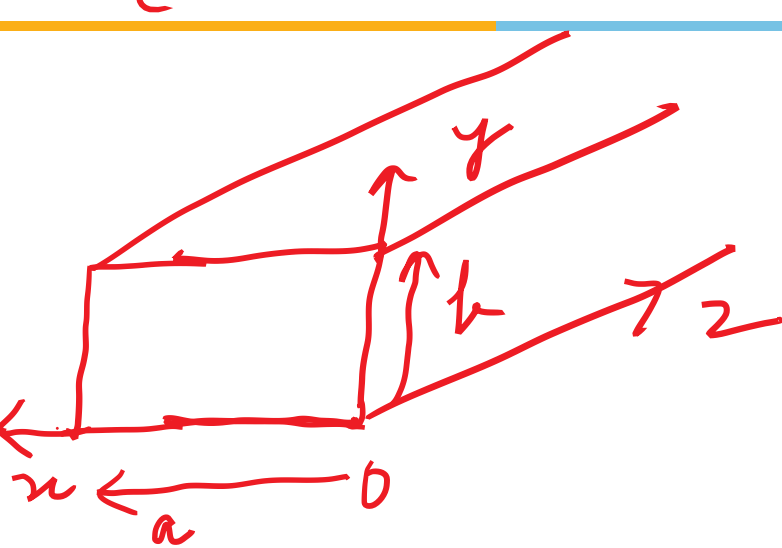
$$E_z = C_5 e^{-j\beta'_z z}$$



$$\sin\left(\frac{m\pi x}{a}\right) (C_3 \cos By + C_4 \sin By)$$

Rectangular Waveguide

(TM)



$$\left[\begin{array}{l} \text{At } y = 0 \quad E_z = 0 \\ \text{means} \\ \underline{C_3 = 0} \end{array} \right]$$

$$\begin{array}{l} \text{At } y = b \quad \underline{E_z = 0} \\ \text{means} \end{array}$$

$$Bb = n\pi$$

$$\boxed{b = \frac{n\pi}{B}}$$

So



$$\left[E_z = C_2 C_4 e^{-j\beta' z} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \right]$$

Rectangular Waveguide



(TM)

$$\frac{1}{x} \frac{\partial^2 x}{\partial x^2} + \frac{1}{y} \frac{\partial^2 y}{\partial y^2} + \frac{1}{z} \frac{\partial^2 z}{\partial z^2} = -\omega^2 \mu \epsilon$$

$$-A^2 - B^2 - \beta'^2 = -\omega^2 \mu \epsilon$$

$$\beta' = \sqrt{\omega^2 \mu \epsilon - A^2 - B^2}$$

$$A = \frac{m\pi}{a}$$

$$B = \frac{n\pi}{b}$$

$$\text{So } \beta' = \sqrt{B^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

$$\beta' = \beta \sqrt{1 - \left(\frac{m\lambda}{2a}\right)^2 - \left(\frac{n\lambda}{2b}\right)^2}$$

Rectangular Waveguide

(TM)



$$\beta' = \beta \sqrt{1 - \left(\frac{m\lambda}{2a}\right)^2 - \left(\frac{n\lambda}{2b}\right)^2}$$

β' should be real

so cut off wavelength λ

$$1 = \left(\frac{m\lambda_c}{2a}\right)^2 + \left(\frac{n\lambda_c}{2b}\right)^2$$

$$\lambda_c = \frac{1}{\sqrt{\left(\frac{m}{2a}\right)^2 + \left(\frac{n}{2b}\right)^2}}$$

cut off wavelength

$$\beta' = \sqrt{\beta^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

Rectangular Waveguide

(TM)



$$\beta' = \sqrt{\omega^2 \mu \epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

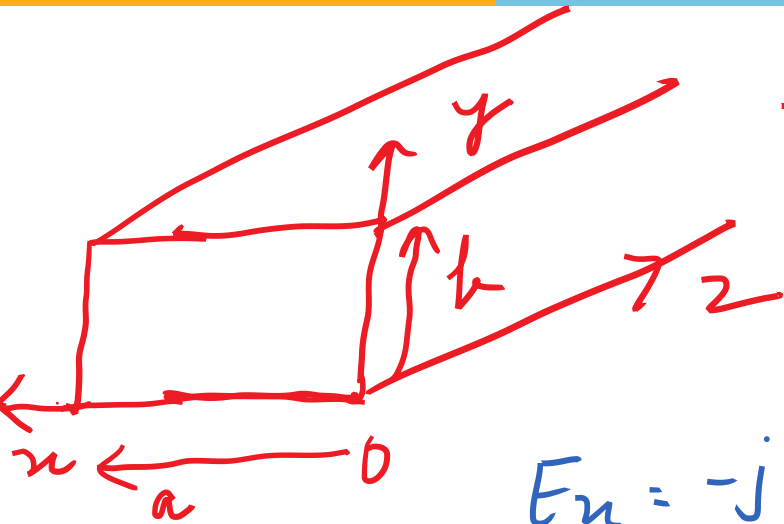
cut off frequency:

$$\omega^2 \mu \epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$\left[\begin{aligned} f_c &= \frac{v_p}{2\pi} \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}} \\ f_c &= \frac{v_p}{2} \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}} \end{aligned} \right]$$

Rectangular Waveguide

(TM)



So $E_z = C e^{-j\beta' z} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)$

$$E_x = -\frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y} - \frac{j\beta'}{h^2} \frac{\partial E_z}{\partial x}$$

where $h = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$

$$E_y = \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x} - \frac{j\beta'}{h^2} \frac{\partial E_z}{\partial y}$$

$$H_x = \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial y} - \frac{j\beta'}{h^2} \frac{\partial H_z}{\partial x} \quad H_y = -\frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial x} - \frac{j\beta'}{h^2} \frac{\partial H_z}{\partial y}$$

Rectangular Waveguide



(TM)

$$E_z = C \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta' z}$$

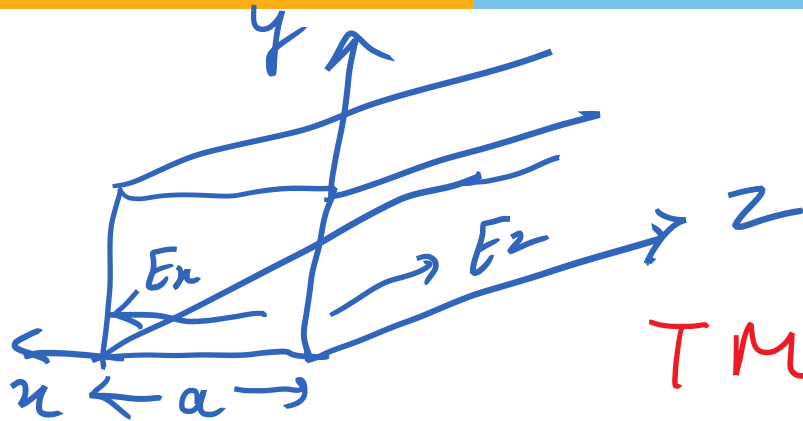
$$E_x = -\frac{j\beta'}{h^2} \left(\frac{m\pi}{a}\right) C \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta' z}$$

$$E_y = -\frac{j\beta'}{h^2} \left(\frac{n\pi}{b}\right) C \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta' z}$$

$$H_x = \frac{j\omega \epsilon}{h^2} \left(\frac{n\pi}{b}\right) C \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta' z}$$

$$H_y = -\frac{j\omega \epsilon}{h^2} \left(\frac{m\pi}{a}\right) C \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta' z}$$

Rectangular Waveguide (TM) *(Existence of modes)*



$$h = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

TM_{mn} modes m → no. of half cycles along x dirⁿ

n → no. of half cycles along y dirⁿ

TM₀₀ X

TM₀₁ X

TM₁₀ X

TM₁₁ ✓

* TM_{m0} X

TM_{n0} X

Smallest
TM₁₁ ✓

Rectangular Waveguide (TE)



$$[E_z = 0]$$

$$\nabla^2 H_z = -\omega^2 \mu \epsilon H_z$$

$$\frac{\partial^2}{\partial x^2} H_z + \frac{\partial^2}{\partial y^2} H_z + \frac{\partial^2}{\partial z^2} H_z = -\omega^2 \mu \epsilon H_z$$

$$H_z = X Y Z$$

$$\frac{1}{x} \frac{\partial^2 X}{\partial x^2} + \frac{1}{y} \frac{\partial^2 Y}{\partial y^2} + \frac{1}{z} \frac{\partial^2 Z}{\partial z^2} = -\omega^2 \mu \epsilon$$

$$\downarrow$$
$$-A^2$$

$$\downarrow$$
$$-B^2$$

$$\downarrow$$
$$-\beta'^2$$

Rectangular Waveguide (TE)



$$X = C_1 \cos Ax + C_2 \sin Ax$$

$$Y = C_3 \cos By + C_4 \sin By$$

$$Z = C_5 e^{-j\beta' z} + C_6 e^{j\beta' z}$$

Applying boundary conditions!

$C_6 = 0$ No backward wave

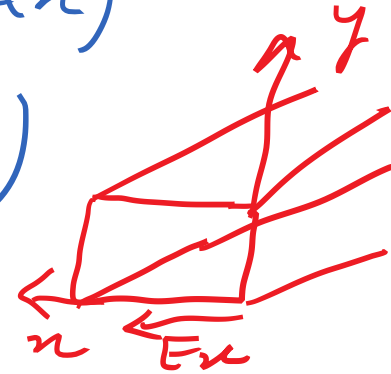
$$H(z) = C_5 e^{-j\beta' z} (C_1 \cos Ax + C_2 \sin Ax) \\ (C_3 \cos By + C_4 \sin By)$$

Rectangular Waveguide (TE)



$$H(z) = C_5 e^{-j\beta' z} (C_1 \cos Ax + C_2 \sin Ax) \\ (C_3 \cos By + C_4 \sin By)$$

$$E_n = -\frac{j\beta'}{h^2} \frac{\partial E_z}{\partial x} - \frac{j\omega \mu}{h^2} \frac{\partial H_z}{\partial y}$$



$$E_n = -\frac{j\omega \mu}{h^2} (C_1 \cos Ax + C_2 \sin Ax)$$

At $y=0$
 $y=b$

$$(-B C_3 \sin By + B C_4 \cos By) E_n = 0$$

Rectangular Waveguide (TE)



At $y=0$ $E_n = 0$

$C_4 = 0$

$$E_n = -\frac{j\omega\mu}{h^2} \left(C_1 \cos An + C_2 \sin An + \left(-\frac{n\pi}{b}\right) C_3 \sin\left(\frac{n\pi}{b} y\right) \right)$$

$$E_n = -\frac{j\omega\mu}{h^2} (C_1 \cos An + C_2 \sin An)$$

At $y=b$ $E_n = 0$

$Bb = n\pi$
 $B = \frac{n\pi}{b}$

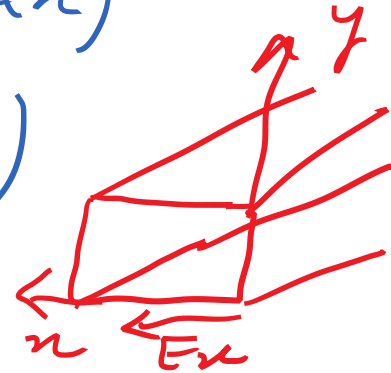
At $y=0$ $E_n = 0$

$$(-B C_3 \sin By + B C_4 \cos By) E_n = 0$$

Rectangular Waveguide (TE)



$$H(z) = C_5 e^{-j\beta' z} (C_1 \cos Ax + C_2 \sin Ax) \\ (C_3 \cos By + C_4 \sin By)$$



$$E_y = -\frac{j\beta'}{h^2} \frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial z}$$

$$E_y = \frac{j\omega\mu}{h^2} (C_5 e^{-j\beta' z} C_3 \cos By (-AC_1 \sin Ax + AC_2 \cos Ax)$$

Rectangular Waveguide (TE)



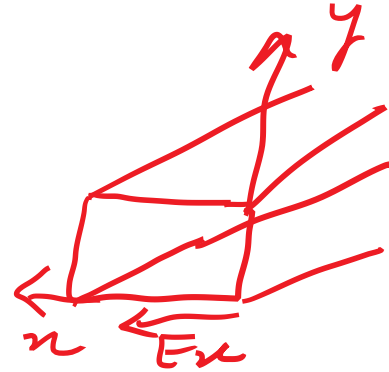
$$\underline{x=0} \quad E_y = 0$$

$$C_2 = 0$$

$$x=a \quad E_y = 0$$

$$Aa = m\bar{\lambda}$$

$$A = \frac{m\bar{\lambda}}{a}$$



$$\text{At } x=0 \quad E_y = 0 \\ x=a \quad E_y = 0$$

$$E_y = \frac{j\omega\mu}{h^2} \left(C_5 e^{-j\beta'z} C_3 \cos By (-AC_1 \sin Ax + AC_2 \cos Ax) \right)$$

Rectangular Waveguide (TE)



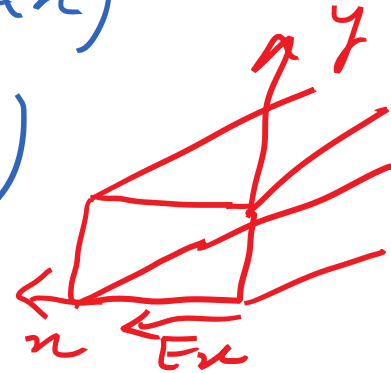
$$H(z) = C_5 e^{-j\beta' z} (C_1 \cos Ax + C_2 \sin Ax) \\ (C_3 \cos By + C_4 \sin By)$$

So $C_4 = 0$

$C_2 = 0$

$A = \frac{m\pi}{a}$

$B = \frac{n\pi}{b}$



So $H(z) = C_1 C_3 C_5 e^{-j\beta' z} \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)$

✓ A $H(z) = C e^{-j\beta' z} \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)$

Rectangular Waveguide (TE)



$$E_x = \frac{j\omega\mu}{h^2} C \left(\frac{n\pi}{b} \right) \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-j\beta'z}$$

$$E_y = -\frac{j\omega\mu}{h^2} C \left(\frac{m\pi}{a} \right) \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-j\beta'z}$$

$$H_x = -\frac{j\beta'}{h^2} C \left(\frac{m\pi}{a} \right) \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-j\beta'z}$$

$$H_y = -\frac{j\beta'}{h^2} C \left(\frac{n\pi}{b} \right) \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-j\beta'z}$$

$$\underline{\underline{H(z) = C e^{-j\beta'z} \cos\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)}}$$

Rectangular Waveguide

(TE) (Existence of modes)



$TE_{00} \longrightarrow \times$

$TE_{01} \longrightarrow \checkmark$

$TE_{10} \longrightarrow \checkmark$

$TE_{11} \longrightarrow \checkmark$

$TE_{m0} \rightarrow$ exists

TE_{0n} exists

$$f_{\text{cutoff}} = \frac{v_p}{2} \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}$$

So frequency for TE & TM modes are same.

Rectangular Waveguide (TE)



$$f_c = \frac{v_p}{2} \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}} \quad a > b$$

$$TE_{01} \rightarrow f_{c_{01}} = \frac{v_p}{2b}$$

$$TE_{10} \rightarrow f_{c_{10}} = \frac{v_p}{2a}$$

$$TE_{11} \rightarrow f_{c_{11}} = \frac{v_p}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$$

TM_{01} is
dominant mode
in parallel
plate
waveguide

TE_{10} is
called

dominant mode
in rectangular

So

$$f_{c_{10}} < f_{c_{01}} < f_{c_{11}}$$

waveguide

Rectangular Waveguide

(TE) $E_z = 0$

E_{10}



$$E_x = \frac{j\omega\mu}{h^2} c \left(\frac{n\pi}{b} \right) \cos \left(\frac{m\pi}{a} x \right) \sin \left(\frac{n\pi}{b} y \right) e^{-j\beta'z}$$

$$E_y = -\frac{j\omega\mu}{h^2} c \left(\frac{m\pi}{a} \right) \sin \left(\frac{m\pi}{a} x \right) \cos \left(\frac{n\pi}{b} y \right) e^{-j\beta'z}$$

$$H_x = -\frac{j\beta'}{h^2} c \left(\frac{m\pi}{a} \right) \sin \left(\frac{m\pi}{a} x \right) \cos \left(\frac{n\pi}{b} y \right) e^{-j\beta'z}$$

$$H_y = -\frac{j\beta'}{h^2} c \left(\frac{n\pi}{b} \right) \cos \left(\frac{m\pi}{a} x \right) \sin \left(\frac{n\pi}{b} y \right) e^{-j\beta'z}$$

$$H(z) = C e^{-j\beta'z} \cos \left(\frac{m\pi}{a} x \right) \cos \left(\frac{n\pi}{b} y \right)$$

Rectangular Waveguide (TE₁₀)



$$\underline{E_x = 0}$$

$$\underline{H_y = 0}$$

$$\underline{E_z = 0}$$

$$\left[\begin{array}{l} E_y = -j C_1 \sin\left(\frac{\pi x}{a}\right) e^{-j\beta' z} \\ H_x = +j C_2 \sin\left(\frac{\pi x}{a}\right) e^{-j\beta' z} \\ H_z = C_3 \cos\left(\frac{\pi x}{a}\right) e^{-j\beta' z} \end{array} \right]$$

$$\left[\begin{array}{l} \text{Re}[E_y] = + C_1 \sin\left(\frac{\pi x}{a}\right) \sin(\beta' z) \\ \text{Re}[H_x] = + C_2 \sin\left(\frac{\pi x}{a}\right) \sin(\beta' z) \end{array} \right] \left[\begin{array}{l} \text{Re}[H_z] = C_3 \cos\left(\frac{\pi x}{a}\right) \\ \cos(\beta' z) \end{array} \right]$$